

Topic 11: Second-Order Linear Equations & The 3 Characteristic Cases

Professor Leonard Master Series · Lessons 37 to 39

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: Why $y = e^{rx}$?

"Why do we guess $y = e^{rx}$ for constant-coefficient equations $ay'' + by' + cy = 0$?"

Professor Leonard explains:

"Think about what this equation says. It says: a multiple of the second derivative, plus a multiple of the first derivative, plus a multiple of the original function must CANCEL OUT to zero for ALL x . What is the ONLY function in mathematics whose derivatives are multiples of itself? The exponential function e^{rx} !"

Substitute $y = e^{rx}$, $y' = re^{rx}$, $y'' = r^2e^{rx}$:

$$a(r^2e^{rx}) + b(re^{rx}) + c(e^{rx}) = 0$$

$$(ar^2 + br + c)e^{rx} = 0$$

Since e^{rx} is never zero, we get the **Characteristic Auxiliary Equation**:

$$ar^2 + br + c = 0$$

2. The Three Great Cases

Case 1: Distinct Real Roots ($r_1 \neq r_2$)

$$y_h(x) = C_1e^{r_1x} + C_2e^{r_2x}$$

Case 2: Repeated Real Roots ($r_1 = r_2 = r$)

"If you write $y = C_1e^{rx} + C_2e^{rx}$, those are NOT independent! That is just $(C_1 + C_2)e^{rx} = Ce^{rx}$."

Using reduction of order, the second independent solution picks up a factor of x :

$$y_h(x) = C_1e^{rx} + C_2xe^{rx}$$

Case 3: Complex Conjugate Roots ($r = \alpha \pm i\beta$)

By Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$:

$$e^{(\alpha \pm i\beta)x} = e^{\alpha x}(\cos(\beta x) \pm i \sin(\beta x))$$

Combining the positive and negative branches gives the pure real general solution:

$$y_h(x) = e^{\alpha x} (C_1 \cos(\beta x) + C_2 \sin(\beta x))$$